Savitzky–Golay Filter Correction for Memory-Timescale Inference

Derivation of the bias factor Φ_{SG} via the G-integral, with fully worked example $m = 2,\ d = 2,\ S = 1$

Contents

This document derives the correction factor Φ_{SG} required when Savitzky–Golay (SG) polynomial derivatives replace naive finite-difference estimators in the overdamped-with-memory (OD+memory) GNN inference pipeline. The central result is

$$\Phi_{\text{SG}} \;=\; \frac{\gamma^*}{\gamma_{\text{eff}}} \;=\; \frac{\langle a_{\text{sg}}[t]\, v_{\text{sg}}[t-S]\rangle}{\langle v_{\text{sg}}[t-S]^2\rangle} \cdot \frac{1}{\gamma_{\text{eff}}} \qquad (\varepsilon \to 0),$$

where $\varepsilon = \Delta t/\tau$. For $m = 2,\ d = 2,\ S = 1$ we show step by step that

$$\boxed{\Phi_{\text{SG}} = \tfrac{107}{210} \approx 0.5095.}$$

1 Physical model and notation

We study a particle obeying the overdamped Langevin equation with a persistent (coloured-noise) active force:

$$\gamma\, \dot{x}(t) = F_{\text{env}}(x) + f_A(t), \tag{1}$$

$$\dot{f}_A(t) = -f_A/\tau + \sigma\xi(t), \tag{2}$$

where γ is drag, F_{env} is the environmental force, f_A is the memory force with timescale τ, σ is noise amplitude, and ξ is unit Gaussian white noise.

Assumption 1 (Overdamped)**.** No inertia; the equation of motion is first-order in position.

Assumption 2 (Ornstein–Uhlenbeck active force)**.** f_A is a stationary Gaussian process with autocovariance $C_{f_A}(s) = \frac{\sigma^2\tau}{2} e^{-|s|/\tau}$.

1.1 Effective underdamped representation

Differentiating (1) and substituting (2) gives a single second-order ODE:

$$\ddot{x} = \gamma_{\text{eff}}\, \dot{x} + F_{\text{env}}(x)/(\gamma\tau) + \text{noise}, \qquad \gamma_{\text{eff}} \equiv F'_{\text{env}}(x)/\gamma - 1/\tau. \tag{3}$$

For a linear trap $F_{\text{env}} = -\kappa x$ the effective drag is the negative constant $\gamma_{\text{eff}} = -\kappa/\gamma - 1/\tau$. Benchmark run ($\kappa = 0.5, \gamma = 1, \tau = 10$): $\gamma_{\text{eff}} = -0.60$.

1.2 Velocity autocovariance

To leading order in $\varepsilon = \Delta t/\tau \to 0$, the velocity $v = \dot{x}$ satisfies $\dot{v} \approx \gamma_{\text{eff}} v$, giving an exponentially decaying stationary autocovariance:

$$C_v(s) \equiv \langle v(t)\, v(t+s)\rangle = \sigma_v^2\, e^{\gamma_{\text{eff}}|s|}, \qquad \sigma_v^2 = \text{Var}(v). \tag{4}$$

Assumption 3 (Small ε)**.** $\varepsilon = \Delta t/\tau \ll 1$. For $\Delta t = 0.1$, $\tau = 10$: $\varepsilon = 0.01$, so finite-ε corrections to Φ_{SG} are $O(\varepsilon^2) \approx 0.01\%$.

Assumption 4 (Stationarity)**.** Trajectory is in steady state; burn-in frames must be discarded.

2 Savitzky–Golay derivative estimators

2.1 Polynomial fitting and what the coefficients mean

Fix a frame t. Write the $2m+1$ surrounding positions as $x[t+k]$ for $k = -m, \ldots, m$. The SG filter fits a degree-d polynomial

$$p(k) = a_0 + a_1 k + a_2 k^2 + \cdots + a_d k^d$$

to the data $\{x[t+k]\}$ by minimising the sum of squared residuals:

$$\min_{a_0,\ldots,a_d} \sum_{k=-m}^{m} \big[x[t+k] - p(k)\big]^2. \tag{5}$$

Once p is found, the velocity and acceleration estimates are the values of p' and p'' at the centre $k = 0$.

Why is the r-th derivative at $k = 0$ equal to $r!\, a_r$? Differentiating the polynomial r times and evaluating at $k = 0$: every term $a_j k^j$ with $j \neq r$ either differentiates to zero (if $j < r$) or retains a factor of k (if $j > r$, so still zero at $k = 0$). The single surviving term is $a_r k^r$, whose r-th derivative at $k = 0$ is $r!\, a_r$. In particular:

$$p'(0) = 1!\, a_1 = a_1, \qquad p''(0) = 2!\, a_2 = 2a_2.$$

The SG derivative estimates are therefore

$$v_{\text{sg}}[t] = \frac{p'(0)}{\Delta t} = \frac{a_1}{\Delta t} = \frac{1}{\Delta t} \sum_{k=-m}^{m} c_k^{(v)}\, x[t+k], \tag{6}$$

$$a_{\text{sg}}[t] = \frac{p''(0)}{\Delta t^2} = \frac{2a_2}{\Delta t^2} = \frac{1}{\Delta t^2} \sum_{k=-m}^{m} c_k^{(a)}\, x[t+k], \tag{7}$$

where $c_k^{(v)}$ are the weights that reconstruct a_1 from the data, and $c_k^{(a)}$ are the weights that reconstruct $2a_2$ from the data. Finding these weights is a standard linear algebra problem (each coefficient is a linear function of the data), and the solution is unique given $2m + 1 > d$ (more points than parameters).

2.2 Explicit coefficients for $m = 2$, $d = 2$

Here $k \in \{-2, -1, 0, 1, 2\}$ and the polynomial is quadratic: $p(k) = a_0 + a_1 k + a_2 k^2$. We need three equations to find a_0, a_1, a_2. Differentiating (5) with respect to each a_j and setting to zero gives one equation per coefficient:

$$\frac{\partial}{\partial a_0}: \quad \sum_k \left[x[t+k] - a_0 - a_1 k - a_2 k^2\right] = 0 \quad \Longrightarrow \quad 5a_0 + a_1 \underbrace{\textstyle\sum_k k}_{=0} + a_2 \sum_k k^2 = \sum_k x[t+k], \tag{8}$$

$$\frac{\partial}{\partial a_1}: \quad \sum_k k\left[x[t+k] - a_0 - a_1 k - a_2 k^2\right] = 0 \quad \Longrightarrow \quad a_0 \underbrace{\textstyle\sum_k k}_{=0} + a_1 \sum_k k^2 + a_2 \underbrace{\textstyle\sum_k k^3}_{=0} = \sum_k k\, x[t+k], \tag{9}$$

$$\frac{\partial}{\partial a_2}: \quad \sum_k k^2\left[x[t+k] - a_0 - a_1 k - a_2 k^2\right] = 0 \quad \Longrightarrow \quad a_0 \sum_k k^2 + a_1 \underbrace{\textstyle\sum_k k^3}_{=0} + a_2 \sum_k k^4 = \sum_k k^2 x[t+k]. \tag{10}$$

The odd power sums $\sum_k k$ and $\sum_k k^3$ vanish because the window is symmetric (k runs from $-m$ to m). This is the key simplification: *the a_1 equation (9) decouples completely from a_0 and a_2.*

First-derivative coefficients $c_k^{(v)}$. Equation (9) alone gives

$$a_1 = \frac{\sum_k k\, x[t+k]}{\sum_k k^2}.$$

For $m = 2$: $\sum_k k^2 = (-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2 = 4 + 1 + 0 + 1 + 4 = 10$, so

$$a_1 = \frac{1}{10} \sum_k k\, x[t+k] \qquad \Longrightarrow \qquad c_k^{(v)} = \frac{k}{10}, \qquad k = -2, -1, 0, 1, 2.$$

Second-derivative coefficients $c_k^{(a)}$. We need $2a_2$. Equations (8) and (10) involve both a_0 and a_2 (with $\sum_k k^2 = 10$ and $\sum_k k^4 = 4^2 + 1^2 + 0 + 1^2 + 4^2 = 2(16 + 1) = 34$):

$$5a_0 + 10a_2 = \sum_k x[t+k], \qquad 10a_0 + 34a_2 = \sum_k k^2 x[t+k].$$

Multiply the first equation by 2 and subtract:

$$(34 - 20)a_2 = \sum_k k^2 x[t+k] - 2\sum_k x[t+k] \quad \Longrightarrow \quad 14a_2 = \sum_k (k^2 - 2)\, x[t+k].$$

The second derivative estimate is therefore

$$2a_2 = \frac{1}{7} \sum_k (k^2 - 2)\, x[t+k].$$

Evaluating $k^2 - 2$ at each k: $k = -2:\ 4 - 2 = 2$; $\ k = -1:\ 1 - 2 = -1$; $\ k = 0:\ 0 - 2 = -2$; $k = 1:\ 1 - 2 = -1$; $\ k = 2:\ 4 - 2 = 2$.

So the second-derivative stencil is

$$c_k^{(a)} = \frac{k^2 - 2}{7}, \qquad k = -2, -1, 0, 1, 2,$$

Table 1: SG stencil coefficients for $m = 2$, $d = 2$.

k	$c_k^{(v)}$	$c_k^{(a)}$
-2	$-2/10$	$+2/7$
-1	$-1/10$	$-1/7$
0	0	$-2/7$
$+1$	$+1/10$	$-1/7$
$+2$	$+2/10$	$+2/7$

giving the values $\{2/7, -1/7, -2/7, -1/7, 2/7\}$.

Quick checks. $\sum_k c_k^{(v)} \cdot k = \sum_k k^2/10 = 1$ (recovers derivative of $x = k$). $\sum_k c_k^{(a)} \cdot k^2 = (2 \cdot 4 - 1 - 0 - 1 + 2 \cdot 4)/7 = 2$ (recovers second derivative of $x = k^2$). $\sum_k c_k^{(v)} = 0$ and $\sum_k c_k^{(a)} = 0$ as required (constants vanish).

2.3 Key symmetry

The velocity coefficients $c_k^{(v)} = k/10$ are *odd*: $c_{-k}^{(v)} = -c_k^{(v)}$. The acceleration coefficients $c_k^{(a)}$ are *even*: $c_{-k}^{(a)} = c_k^{(a)}$. This parity structure drives everything that follows.

2.4 The velocity shift S and parity degeneracy

Both $v_{\text{sg}}[t]$ and $a_{\text{sg}}[t]$ are centred on the same frame t. For any stationary process,

$$\langle a_{\text{sg}}[t]\, v_{\text{sg}}[t] \rangle = 0, \tag{11}$$

because $c_k^{(a)}$ is even, $c_l^{(v)}$ is odd, and their product $c_k^{(a)} c_l^{(v)}$ is odd under $(k, l) \to (-k, -l)$, while the correlator $\langle x[t + k]x[t + l] \rangle = C_x(k - l)$ is symmetric under that swap.

The fix is to pair $a_{\text{sg}}[t]$ with a velocity at a *shifted* frame:

$$a_{\text{sg}}[t] \text{ (centred on } t) \quad \text{paired with} \quad v_{\text{sg}}[t - S] \text{ (centred on } t - S).$$

The shift $S \geq 1$ breaks the parity and yields a non-zero, informative correlation.

Assumption 5 (Shift $S \geq 1$)**.** $S = 0$ is identically degenerate. The recommended default is $S = 1$.

Valid frame indices require both windows to fit inside the trajectory: $t \in [m + S,\ n_{\text{frames}} - 1 - m]$. For $m = 2,\ S = 1$: $t \in [3, n - 3]$, yielding $n - 5$ valid frames.

3 The G-integral and the bias factor Φ_{SG}

3.1 Representing positions as velocity integrals

Every position in (6)–(7) can be written as an integral of velocity. Because $\sum_k c_k^{(v)} = \sum_k c_k^{(a)} = 0$, the constant $x[t]$ cancels and only *increments* survive:

$$x[t + k] - x[t] = \operatorname{sgn}(k) \int_0^{|k|\Delta t} v\big(t + \operatorname{sgn}(k)\, u\big)\, du, \qquad k \neq 0. \tag{12}$$

To see this explicitly: writing $x[t+k] = x[t] + (x[t+k] - x[t])$ inside (6) gives $v_{\rm sg}[t] = \frac{1}{\Delta t}\big[x[t]\sum_k c_k^{(v)} + \sum_k c_k^{(v)}(x[t+k]-x[t])\big]$; the first sum vanishes by the stencil normalisation $\sum_k c_k^{(v)} = 0$, leaving only the increments, as in (12). The same argument applies to $a_{\rm sg}$.

Substituting into (6) and (7) (with S-shifted velocity):

$$v_{\rm sg}[t-S] = \frac{1}{\Delta t}\sum_{l\neq 0} c_l^{(v)}\,\operatorname{sgn}(l)\int_0^{|l|\Delta t} v\big((t-S)+\operatorname{sgn}(l)\,w\big)\,dw, \tag{13}$$

$$a_{\rm sg}[t] = \frac{1}{\Delta t^2}\sum_{k\neq 0} c_k^{(a)}\,\operatorname{sgn}(k)\int_0^{|k|\Delta t} v\big(t+\operatorname{sgn}(k)\,u\big)\,du. \tag{14}$$

3.2 The G-integral

Multiplying (14) by (13) and taking expectation, we use the velocity autocovariance $C_v(s) = \langle v(t')v(t'+s)\rangle$:

$$G(k,l,S) \equiv \operatorname{sgn}(k)\,\operatorname{sgn}(l)\int_0^{|k|\Delta t}\int_0^{|l|\Delta t} C_v\big(S\Delta t + \operatorname{sgn}(k)\,u - \operatorname{sgn}(l)\,w\big)\,du\,dw. \tag{15}$$

Then

$$N \equiv \langle a_{\rm sg}[t]\,v_{\rm sg}[t-S]\rangle = \frac{1}{\Delta t^3}\sum_{k\neq 0,\,l\neq 0} c_k^{(a)}c_l^{(v)}\,G(k,l,S), \qquad D \equiv \langle v_{\rm sg}[t-S]^2\rangle = \frac{1}{\Delta t^2}\sum_{k\neq 0,\,l\neq 0} c_k^{(v)}c_l^{(v)}\,G(k,l,0). \tag{16}$$

3.3 The $\varepsilon \to 0$ expansion

Rescale $u = \Delta t\,\tilde u$, $w = \Delta t\,\tilde w$, and expand $C_v(s) = \sigma_v^2 e^{\gamma_{\rm eff}|s|}$ for small $\varepsilon = \gamma_{\rm eff}\Delta t$:

$$G(k,l,S) = \sigma_v^2\,\Delta t^2\;\operatorname{sgn}(k)\,\operatorname{sgn}(l)\int_0^{|k|}\int_0^{|l|}\Big[1+\underbrace{\varepsilon\,|S+\operatorname{sgn}(k)\,\tilde u - \operatorname{sgn}(l)\,\tilde w|}_{O(\varepsilon)} + O(\varepsilon^2)\Big]\,d\tilde u\,d\tilde w. \tag{17}$$

Define the dimensionless double integral

$$I(k,l,S) \equiv \int_0^{|k|}\int_0^{|l|}\big|S+\operatorname{sgn}(k)\,\tilde u - \operatorname{sgn}(l)\,\tilde w\big|\,d\tilde u\,d\tilde w, \tag{18}$$

so that $G(k,l,S) = \sigma_v^2\Delta t^2\big[\operatorname{sgn}(k)\operatorname{sgn}(l)|k||l| + \varepsilon\,I(k,l,S) + O(\varepsilon^2)\big]$.

3.4 Denominator at leading order

Inserting the $O(1)$ term into the denominator sum in (16):

$$D^{(0)} = \sigma_v^2\left(\sum_{k\neq 0} c_k^{(v)}\,\operatorname{sgn}(k)\,|k|\right)^2 = \sigma_v^2\left(\sum_{k\neq 0}\frac{k^2}{10}\right)^2 = \sigma_v^2\cdot 1^2 = \sigma_v^2. \tag{19}$$

(We used $c_k^{(v)} = k/10$, so $c_k^{(v)}\operatorname{sgn}(k)|k| = k^2/10$ and $\sum_k k^2/10 = (4+1+0+1+4)/10 = 1$.)

3.5 Numerator: $O(1)$ term vanishes (parity)

The $O(1)$ contribution to N factors as

$$N^{(0)} = \frac{\sigma_v^2}{\Delta t} \underbrace{\left(\sum_{k \neq 0} c_k^{(a)} \operatorname{sgn}(k)|k| \right)}_{=0} \left(\sum_{l \neq 0} c_l^{(v)} \operatorname{sgn}(l)|l| \right).$$

The vanishing factor: $c_k^{(a)}$ is even and $\operatorname{sgn}(k)|k| = k$ is odd, so each pair $(k, -k)$ cancels:

$$\sum_{k \neq 0} c_k^{(a)} \cdot k = \frac{(-2)(2) + (-1)(-1) + (-1)(0) + (+2)(1) + (+2)(2)}{7} = \frac{-4 + 1 + 0 - 1 + 4}{7} = 0.$$

Hence $N^{(0)} = 0$, confirming (11) for any S.

3.6 Numerator: $O(\varepsilon)$ term gives Φ_{SG}

The first-order contribution is

$$N^{(1)} = \frac{\sigma_v^2 \varepsilon}{\Delta t} \sum_{k \neq 0} \sum_{l \neq 0} c_k^{(a)} c_l^{(v)} \operatorname{sgn}(k) \operatorname{sgn}(l)\, I(k, l, S). \tag{20}$$

Combining with $D^{(0)} = \sigma_v^2$ and recalling $\varepsilon = \gamma_{\mathrm{eff}} \Delta t$:

$$\gamma^* \equiv \frac{N}{D} = \gamma_{\mathrm{eff}} \sum_{k \neq 0} \sum_{l \neq 0} c_k^{(a)} c_l^{(v)} \operatorname{sgn}(k) \operatorname{sgn}(l)\, I(k, l, S) + O(\varepsilon^2).$$

Dividing through by γ_{eff}:

$$\boxed{\Phi_{\mathrm{SG}} = \frac{\gamma^*}{\gamma_{\mathrm{eff}}} = \sum_{k \neq 0, l \neq 0} c_k^{(a)} c_l^{(v)} \operatorname{sgn}(k) \operatorname{sgn}(l)\, I(k, l, S) + O(\varepsilon^2).} \tag{21}$$

This depends only on the SG stencil coefficients and the shift S, and is independent of σ_v^2, γ_{eff}, and Δt individually.

4 Validity of the key assumptions

This section gives explicit derivations showing that the three key approximations—the exponential velocity autocovariance, the vanishing cross-correlation $\langle a_{\mathrm{sg}} \cdot F_{\mathrm{env}} \rangle$, and the neglect of $O(\varepsilon)$ denominator corrections—are all controlled by the same small parameter $\varepsilon = \Delta t / \tau$.

4.1 Exponential velocity autocovariance

Derivation. Differentiating (1) with respect to time and substituting (2):

$$\gamma \dot{v} = F'_{\mathrm{env}}(x)\, v + \dot{f}_A = F'_{\mathrm{env}}(x)\, v - \frac{f_A}{\tau} + \sigma \xi.$$

Using (1) to eliminate $f_A = \gamma v - F_{\rm env}(x)$:

$$\dot{v} = \underbrace{\left(\frac{F'_{\rm env}(x)}{\gamma} - \frac{1}{\tau}\right)}_{\gamma_{\rm eff}(x)} v + \frac{F_{\rm env}(x)}{\gamma\tau} + \frac{\sigma\xi}{\gamma}. \tag{22}$$

Multiplying (22) at time t by $v(t+s)$ and taking expectations, the noise term contributes only at $s = 0$. The position-drift term $F_{\rm env}(x)/(\gamma\tau)$ introduces a correction of relative magnitude

$$\frac{|F_{\rm env}(x)|/(\gamma\tau)\cdot\sigma_v}{|\gamma_{\rm eff}|\,\sigma_v^2} \;\sim\; \frac{\kappa\,\sigma_x}{\gamma\tau\,|\gamma_{\rm eff}|\,\sigma_v} \;=\; O(\varepsilon), \tag{23}$$

where we used $\sigma_v \sim \sigma_x/\tau$ (the velocity scale is the position scale divided by the memory timescale). Dropping the $O(\varepsilon)$ drift term, the velocity satisfies $\dot{v} \approx \gamma_{\rm eff} v$, whose stationary autocovariance is

$$C_v(s) = \sigma_v^2\, e^{\gamma_{\rm eff}|s|},$$

confirming (4) to $O(\varepsilon)$ accuracy.

Local vs. global linearity. The effective damping $\gamma_{\rm eff}(x) = F'_{\rm env}(x)/\gamma - 1/\tau$ depends on position through the local slope of $F_{\rm env}$. For the single-exponential form (4) to hold, $\gamma_{\rm eff}$ must be approximately constant over the range of positions the particle visits *within a single SG window*. The window spans $m\Delta t$ in time, so the relevant displacement is

$$\Delta x_{\rm window} \;\sim\; \sigma_v \cdot m\Delta t = \sigma_v\tau \cdot m\varepsilon.$$

For fixed m this is an $O(\varepsilon)$ fraction of the full displacement range $\sigma_x \sim \sigma_v\tau$. Linearity is therefore a *local* condition: $F'_{\rm env}(x)$ need not be constant over the whole dataset, only over the small neighbourhood explored within one window. Even a strongly nonlinear trap satisfies this provided $\varepsilon \ll 1$.

4.2 Vanishing cross-correlation $\langle a_{\rm sg}[t] \cdot F_{\rm env}(x[t])\rangle \approx 0$

In the OLS fixed-point (24), the gradient of the loss with respect to $1/\tau_{\rm raw}$ contains the cross-term $\langle a_{\rm sg}[t] \cdot F_{\rm env}(x[t])\rangle$. From (22), at leading order $a_{\rm sg}[t] \approx \gamma_{\rm eff}\, v_{\rm sg}[t]$, so this cross-term is proportional to $\langle v_{\rm sg}[t] \cdot F_{\rm env}(x[t])\rangle$. While velocity and force at the *same* time are correlated through the equation of motion, the OLS pairing uses $v_{\rm sg}[t-S]$ at the *shifted* frame. The dynamical correlation between position at t and velocity at $t-S$ decays over the lag $S\Delta t$:

$$\langle F_{\rm env}(x(t))\, v(t - S\Delta t)\rangle \;\approx\; -\kappa\langle x(t)\, v(t-S\Delta t)\rangle \;\sim\; \kappa\,\sigma_x\,\sigma_v \cdot |\gamma_{\rm eff}|\, S\Delta t,$$

while the leading numerator term scales as $\langle a_{\rm sg}\, v_{\rm sg}\rangle \sim |\gamma_{\rm eff}|\,\sigma_v^2$. Their ratio is

$$\frac{\langle F_{\rm env}(x[t]) \cdot v_{\rm sg}[t-S]\rangle}{\langle a_{\rm sg}[t] \cdot v_{\rm sg}[t-S]\rangle} \;=\; O\!\left(\frac{\kappa\,\sigma_x\, S\Delta t}{\sigma_v}\right) \;=\; O(S\varepsilon),$$

using $\sigma_x \sim \sigma_v\tau$. For fixed S and $\varepsilon \ll 1$ this is $O(\varepsilon)$, so $\langle a_{\rm sg} \cdot F_{\rm env}\rangle \approx 0$ is a controlled approximation to the same order as all other corrections.

4.3 Neglecting $O(\varepsilon)$ denominator corrections

Expanding both N and D to first order in ε:

$$\begin{aligned} N &= N^{(0)} + \varepsilon\, N^{(1)} + O(\varepsilon^2) = \varepsilon\, N^{(1)} + O(\varepsilon^2),\\ D &= D^{(0)} + \varepsilon\, D^{(1)} + O(\varepsilon^2) = \sigma_v^2 + \varepsilon\, D^{(1)} + O(\varepsilon^2), \end{aligned}$$

where $N^{(0)} = 0$ by parity (Section 3.5) and $D^{(0)} = \sigma_v^2 \neq 0$ (Section 3.4). Forming the ratio:

$$\begin{aligned} R \;=\; \frac{N}{D} &= \frac{\varepsilon\, N^{(1)} + O(\varepsilon^2)}{\sigma_v^2\big(1 + \varepsilon\, D^{(1)}/\sigma_v^2 + O(\varepsilon^2)\big)} \\ &= \frac{\varepsilon\, N^{(1)}}{\sigma_v^2}\left(1 - \varepsilon\,\frac{D^{(1)}}{\sigma_v^2} + O(\varepsilon^2)\right) \\ &= \frac{\varepsilon\, N^{(1)}}{\sigma_v^2} + O(\varepsilon^2). \end{aligned}$$

The $O(\varepsilon)$ correction to D enters R only at $O(\varepsilon^2)$, one full power of ε below the leading term. The reason is that $D^{(0)} \neq 0$: the $\varepsilon D^{(1)}$ shift is a *relative* correction to a non-zero baseline, so its effect on the ratio is suppressed by one further factor of ε. Dropping the denominator correction $D^{(\varepsilon)}$ is therefore consistent with the $O(\varepsilon^2)$ accuracy maintained throughout.

5 The corrected OLS fixed-point equation

5.1 GNN prediction and loss gradient

The GNN predicts the acceleration as

$$a_{\text{pred}}[t] = -\frac{v_{\text{sg}}[t-S]}{\tau_{\text{raw}}} + \frac{F_{\text{env}}(x[t])}{\gamma\, \tau_{\text{raw}}} + (\text{pairwise interaction terms}).$$

Setting $\partial\mathcal{L}/\partial\tau_{\text{raw}} = 0$ in the MSE loss $\mathcal{L} = \langle (a_{\text{sg}} - a_{\text{pred}})^2\rangle$, and using that $\langle a_{\text{sg}} \cdot F_{\text{env}}(x)\rangle \approx 0$ (Section 4.2), the fixed-point condition is

$$R_{\text{data}} \equiv \frac{\langle a_{\text{sg}}[t]\, v_{\text{sg}}[t-S]\rangle}{\langle v_{\text{sg}}[t-S]^2\rangle} = \Phi_{\text{SG}} \cdot \gamma_{\text{eff}}. \tag{24}$$

5.2 Two roots and the correct branch

Substituting $\gamma_{\text{eff}} = F'_{\text{env}}/\gamma - 1/\tau$ for a linear trap ($F'_{\text{env}} = -\kappa$):

$$R_{\text{data}} = \Phi_{\text{SG}}\left(-\frac{\kappa}{\gamma} - \frac{1}{\tau}\right).$$

This is a quadratic in τ with generically two real roots. Benchmark ($\kappa = 0.5, \gamma = 1, \tau = 10$):

$$R_{\text{data}}^{\text{theory}} = \frac{107}{210} \times (-0.60) \approx -0.306.$$

The two roots are $\tau \approx 10$ (correct) and $\tau \approx 2$ (spurious). Training converges to the correct root when:

1. R_{data} is accurate (sufficient frames; correct per-frame normalization – no discontinuities at split boundaries).

2. τ_{init} is on the correct branch (default 5.0 is closer to 10 than to the spurious root).

3. $\varepsilon = \Delta t/\tau$ is small.

6 Naive finite-difference estimators via the G-integral

Having established the G-integral machinery and the formula (21), we apply it to the naive finite-difference estimators and recover the 2/3 factor without any separate calculation.

6.1 Stencil coefficients and parity

The naive scheme uses a backward-difference velocity paired with a central-difference acceleration at the same frame index ($S = 0$):

$$v_{\text{naive}}[t] = \frac{x[t] - x[t-1]}{\Delta t}, \tag{25}$$

$$a_{\text{naive}}[t] = \frac{x[t+1] - 2x[t] + x[t-1]}{\Delta t^2}. \tag{26}$$

Assumption 6 (Same-time pairing)**.** $a_{\text{naive}}[t]$ is paired with $v_{\text{naive}}[t]$ at the same frame index ($S = 0$).

Both estimators are linear in positions with stencil sums $\sum_k c_k^{(v)} = \sum_k c_k^{(a)} = 0$, so the increment representation (12) applies exactly as for SG. After factoring out the constant via $\sum_k c_k^{(r)} = 0$, the support reduces to $k \neq 0$:

Table 2: Naive stencil coefficients (non-zero k only after factoring).

k	$c_k^{(v)}$	$c_k^{(a)}$
-1	-1	$+1$
$+1$	—	$+1$

The acceleration stencil $c_k^{(a)}$ is *even*: $c_{+1}^{(a)} = c_{-1}^{(a)} = 1$. The parity sum therefore vanishes,

$$\sum_{k\neq 0} c_k^{(a)} \cdot \operatorname{sgn}(k) \cdot |k| = (+1)(+1) + (+1)(-1) = 0,$$

so $N^{(0)} = 0$ by the same parity argument as Section 3.5, even at $S = 0$. The denominator leading term follows from (19):

$$D^{(0)} = \sigma_v^2 \left(\sum_{l\neq 0} c_l^{(v)} \operatorname{sgn}(l)\, |l| \right)^2 = \sigma_v^2 \big[(-1)(-1)(1)\big]^2 = \sigma_v^2.$$

6.2 The two contributing I-integrals

With $S = 0$, formula (21) sums over pairs $(k, l) \in \{+1, -1\} \times \{-1\}$. The I-integral (18) at $S = 0$ reduces to

$$I(k, l, 0) = \int_0^{|k|} \int_0^{|l|} \big|\operatorname{sgn}(k)\, \tilde{u} - \operatorname{sgn}(l)\, \tilde{w}\big|\, d\tilde{u}\, d\tilde{w}.$$

$I(+1, -1, 0)$: signs $(+, -)$, always separated. The integrand is $\tilde{u} + \tilde{w}$ (positive for all $\tilde{u}, \tilde{w} \geq 0$):

$$I(+1, -1, 0) = \int_0^1 \int_0^1 (\tilde{u} + \tilde{w})\, d\tilde{u}\, d\tilde{w} = \int_0^1 \left(\tfrac{1}{2} + \tilde{w}\right) d\tilde{w} = 1.$$

$I(-1,-1,0)$**: signs** $(-,-)$**, partially overlapping.** The integrand is $|\tilde{w}-\tilde{u}|$; by symmetry:

$$I(-1,-1,0) = \int_0^1\int_0^1 |\tilde{w}-\tilde{u}|\,d\tilde{u}\,d\tilde{w} = 2\int_0^1\int_0^{\tilde{w}} (\tilde{w}-\tilde{u})\,d\tilde{u}\,d\tilde{w} = 2\int_0^1 \frac{\tilde{w}^2}{2}\,d\tilde{w} = \frac{1}{3}.$$

6.3 Assembling $\Phi_{\text{naive}} = 2/3$

Substituting into (21) with $S=0$:

$$\begin{aligned}
\Phi_{\text{naive}} &= \sum_{k\neq 0}\sum_{l\neq 0} c_k^{(a)}\, c_l^{(v)}\ \text{sgn}(k)\ \text{sgn}(l)\, I(k,l,0) \\
&= \underbrace{c_{+1}^{(a)}\, c_{-1}^{(v)}\ \text{sgn}(+1)\ \text{sgn}(-1)}_{(+1)(-1)(+1)(-1)=+1} I(+1,-1,0)\ +\ \underbrace{c_{-1}^{(a)}\, c_{-1}^{(v)}\ \text{sgn}(-1)\ \text{sgn}(-1)}_{(+1)(-1)(-1)(-1)=-1} I(-1,-1,0) \\
&= (+1)(1)\ +\ (-1)\left(\tfrac{1}{3}\right) = 1-\tfrac{1}{3}.
\end{aligned}$$

$$\boxed{R_{\text{naive}} = \frac{\langle a_{\text{naive}}[t]\, v_{\text{naive}}[t]\rangle}{\langle v_{\text{naive}}[t]^2\rangle} = \Phi_{\text{naive}}\cdot\gamma_{\text{eff}} = \frac{2}{3}\,\gamma_{\text{eff}}.}$$

The two integrals have a direct interpretation. $I(+1,-1,0)=1$: with $S=0$ the acceleration window reaches *forward* ($k=+1$) while the velocity window reaches *backward* ($l=-1$), so the two integration regions lie on opposite sides of t and the lag $\tilde{u}+\tilde{w}$ is always positive—full correlation weight. $I(-1,-1,0)=\frac{1}{3}$: both windows reach *backward* ($k=l=-1$), so they partially overlap and the lag $|\tilde{w}-\tilde{u}|$ is sometimes zero—reduced weight $\frac{1}{3}$. Their difference $1-\frac{1}{3}=\frac{2}{3}$ is the bias factor.

7 Worked example: $m=2$, $d=2$, $S=1$

We evaluate (21) completely for $m=2$, $d=2$, $S=1$.

7.1 Reformulation in terms of signed coefficients

It is convenient to define

$$\tilde{c}_k^a \equiv c_k^{(a)}\cdot\text{sgn}(k), \qquad \tilde{c}_l^v \equiv c_l^{(v)}\cdot\text{sgn}(l) = |l|/10, \tag{27}$$

so that (21) becomes $\Phi_{\text{SG}} = \sum_{k,l}\tilde{c}_k^a\,\tilde{c}_l^v\, I(k,l,1)$. The signed acceleration coefficients are:

$k:$	-2	-1	$+1$	$+2$
$\tilde{c}_k^a:$	$-2/7$	$+1/7$	$-1/7$	$+2/7$

Note $\tilde{c}_k^a$ is *odd* ($\tilde{c}_{-k}^a = -\tilde{c}_k^a$), while $\tilde{c}_l^v = |l|/10$ is always non-negative.

7.2 The integral $I(k,l,1)$

From (18) with $S=1$, writing $s_k=\text{sgn}(k)$, $s_l=\text{sgn}(l)$, $a=|k|$, $b=|l|$:

$$I(k,l,1) = \int_0^a\int_0^b |1+s_k\,\tilde{u}-s_l\,\tilde{w}|\,d\tilde{u}\,d\tilde{w}.$$

There are four sign-pattern classes.

7.2.1 Class A: $(s_k, s_l) = (+,+)$

$A(a,b) = \int_0^a \int_0^b |1+\tilde{u}-\tilde{w}|\, d\tilde{u}\, d\tilde{w}$. The integrand changes sign where $\tilde{w} = 1+\tilde{u}$.

$A(1,1)$**.** Since $b = 1 \le 1+\tilde{u}$ for all $\tilde{u} \ge 0$, the integrand is everywhere positive:

$$A(1,1) = \int_0^1 \int_0^1 (1+\tilde{u}-\tilde{w})\, d\tilde{w}\, d\tilde{u} = \int_0^1 \Big(1+\tilde{u}-\tfrac{1}{2}\Big) d\tilde{u} = \int_0^1 \Big(\tfrac{1}{2}+\tilde{u}\Big) d\tilde{u} = \tfrac{1}{2}+\tfrac{1}{2} = 1.$$

$A(1,2)$**.** For $\tilde{u} \in [0,1]$ and $b = 2$, the zero $\tilde{w} = 1+\tilde{u} \in [1,2]$ always lies within $[0,2]$. Splitting at the zero for each fixed $\tilde{u}$:

$$\int_0^2 |1+\tilde{u}-\tilde{w}|\, d\tilde{w} = \int_0^{1+\tilde{u}} (1+\tilde{u}-\tilde{w})\, d\tilde{w} + \int_{1+\tilde{u}}^2 (\tilde{w}-1-\tilde{u})\, d\tilde{w} = \frac{(1+\tilde{u})^2}{2} + \frac{(1-\tilde{u})^2}{2}.$$

Hence:

$$A(1,2) = \int_0^1 \frac{(1+\tilde{u})^2+(1-\tilde{u})^2}{2}\, d\tilde{u} = \int_0^1 (1+\tilde{u}^2)\, d\tilde{u} = 1+\tfrac{1}{3} = \tfrac{4}{3}.$$

$A(2,1)$**.** For $b = 1$, the zero $\tilde{w} = 1+\tilde{u} \ge 1 = b$ for all $\tilde{u} \ge 0$, so the integrand is everywhere positive:

$$A(2,1) = \int_0^2 \int_0^1 (1+\tilde{u}-\tilde{w})\, d\tilde{w}\, d\tilde{u} = \int_0^2 \Big(\tfrac{1}{2}+\tilde{u}\Big) d\tilde{u} = 1+2 = 3.$$

$A(2,2)$**.** The zero $\tilde{w} = 1+\tilde{u}$ lies inside $[0,2]^2$ for $\tilde{u} \in [0,1]$:

$$\int_0^2 |1+\tilde{u}-\tilde{w}|\, d\tilde{u} = \begin{cases} 4-2\tilde{w} & \text{inner over } \tilde{w},\ \tilde{w} \in [0,1] \\ \dfrac{(\tilde{w}-1)^2+(3-\tilde{w})^2}{2} & \tilde{w} \in [1,2] \end{cases}$$

Let me integrate over $\tilde{w}$ as the outer variable: for $\tilde{w} \in [0,1]$: $1+\tilde{u} > \tilde{w}$ for all $\tilde{u} \ge 0$, so integrating over $\tilde{u} \in [0,2]$ gives $\int_0^2 (1+\tilde{u}-\tilde{w}) d\tilde{u} = 4-2\tilde{w}$. For $\tilde{w} \in [1,2]$: zero at $\tilde{u} = \tilde{w}-1$: $\int_0^2 |1+\tilde{u}-\tilde{w}| d\tilde{u} = \frac{(\tilde{w}-1)^2}{2} + \frac{(3-\tilde{w})^2}{2}$.

$$A(2,2) = \int_0^1 (4-2\tilde{w})\, d\tilde{w} + \int_1^2 \frac{(\tilde{w}-1)^2+(3-\tilde{w})^2}{2}\, d\tilde{w} = 3+\frac{1}{2}\cdot\frac{8}{3} = 3+\frac{4}{3} = \frac{13}{3}.$$

7.2.2 Class B: $(s_k, s_l) = (+,-)$

$B(a,b) = \int_0^a \int_0^b (1+\tilde{u}+\tilde{w})\, d\tilde{u}\, d\tilde{w} = ab(1+a/2+b/2)$, always positive.

$$B(1,1) = 2, \quad B(1,2) = 5, \quad B(2,1) = 5, \quad B(2,2) = 12.$$

7.2.3 Class C: $(s_k, s_l) = (-,+)$

$C(a,b) = \int_0^a \int_0^b |1-\tilde{u}-\tilde{w}|\, d\tilde{u}\, d\tilde{w}$. Zero at $\tilde{u}+\tilde{w} = 1$.

$C(1,1)$**.** Zero line $\tilde{u}+\tilde{w} = 1$ cuts through $[0,1]^2$. For fixed $\tilde{u}$, split $\tilde{w}$ at $1-\tilde{u}$:

$$\int_0^1 |1-\tilde{u}-\tilde{w}|\, d\tilde{w} = \int_0^{1-\tilde{u}} (1-\tilde{u}-\tilde{w})\, d\tilde{w} + \int_{1-\tilde{u}}^1 (\tilde{u}+\tilde{w}-1)\, d\tilde{w} = \frac{(1-\tilde{u})^2}{2} + \frac{\tilde{u}^2}{2}.$$

$$C(1,1) = \int_0^1 \frac{(1-\tilde{u})^2 + \tilde{u}^2}{2}\, d\tilde{u} = \frac{1}{3}.$$

$C(1,2)$**.** Integrate $\tilde{u}$ first, outer $\tilde{w} \in [0,2]$:

- $\tilde{w} \in [0,1]$: zero at $\tilde{u} = 1 - \tilde{w} \in [0,1]$, giving $\int_0^1 |1 - \tilde{u} - \tilde{w}|\, d\tilde{u} = \frac{(1-\tilde{w})^2}{2} + \frac{\tilde{w}^2}{2}$.
- $\tilde{w} \in [1,2]$: $1 - \tilde{u} - \tilde{w} < 0$ for all $\tilde{u} \geq 0$, giving $\int_0^1 (\tilde{u} + \tilde{w} - 1)\, d\tilde{u} = \tilde{w} - \frac{1}{2}$.

$$C(1,2) = \int_0^1 \frac{(1-\tilde{w})^2 + \tilde{w}^2}{2}\, d\tilde{w} + \int_1^2 \Big(\tilde{w} - \tfrac{1}{2}\Big) d\tilde{w} = \frac{1}{3} + 1 = \frac{4}{3}.$$

$C(2,1)$**.** Outer $\tilde{w} \in [0,1]$, inner $\tilde{u} \in [0,2]$; zero at $\tilde{u} = 1 - \tilde{w} \in [0,1]$:

$$\int_0^2 |1 - \tilde{u} - \tilde{w}|\, d\tilde{u} = \frac{(1-\tilde{w})^2}{2} + \frac{(1+\tilde{w})^2}{2}.$$

$$C(2,1) = \int_0^1 \frac{(1-\tilde{w})^2 + (1+\tilde{w})^2}{2}\, d\tilde{w} = \int_0^1 (1 + \tilde{w}^2)\, d\tilde{w} = \frac{4}{3}.$$

$C(2,2)$**.** Outer $\tilde{w} \in [0,2]$:

- $\tilde{w} \in [0,1]$: zero at $\tilde{u} = 1 - \tilde{w} \in [0,1]$, $\int_0^2 |1 - \tilde{u} - \tilde{w}|\, d\tilde{u} = \frac{(1-\tilde{w})^2}{2} + \frac{(1+\tilde{w})^2}{2}$.
- $\tilde{w} \in [1,2]$: always negative, $\int_0^2 (\tilde{u} + \tilde{w} - 1)\, d\tilde{u} = 2\tilde{w}$.

$$C(2,2) = \int_0^1 (1 + \tilde{w}^2)\, d\tilde{w} + \int_1^2 2\tilde{w}\, d\tilde{w} = \frac{4}{3} + 3 = \frac{13}{3}.$$

7.2.4 Class D: $(s_k, s_l) = (-,-)$

$D(a,b) = \int_0^a \int_0^b |1 - \tilde{u} + \tilde{w}|\, d\tilde{u}\, d\tilde{w}$. Swapping $\tilde{u} \leftrightarrow \tilde{w}$ gives $D(a,b) = A(b,a)$ (class A with roles of a and b exchanged):

$D(1,1) = A(1,1) = 1, \quad D(2,1) = A(1,2) = \frac{4}{3}, \quad D(1,2) = A(2,1) = 3, \quad D(2,2) = A(2,2) = \frac{13}{3}.$

7.2.5 Summary table

7.3 Assembling Φ_{SG} via the H_k sums

It is efficient to first sum over l for each fixed k, defining $H_k = \sum_{l \neq 0} \tilde{c}_l^v\, I(k,l,1)$. Recall $\tilde{c}_l^v = |l|/10$.

$k = +1$**:** using I values from rows 1–4 of Table 3,

$$H_{+1} = \frac{1}{10}(1) + \frac{1}{10}(2) + \frac{2}{10}\cdot\frac{4}{3} + \frac{2}{10}(5) = \frac{1+2}{10} + \frac{8}{30} + 1 = \frac{9+8}{30} + 1 = \frac{17}{30} + 1 = \frac{47}{30}.$$

$k = -1$**:** rows 5–8,

$$H_{-1} = \frac{1}{10}\cdot\frac{1}{3} + \frac{1}{10}(1) + \frac{2}{10}\cdot\frac{4}{3} + \frac{2}{10}(3) = \frac{1}{30} + \frac{3}{30} + \frac{8}{30} + \frac{18}{30} = \frac{30}{30} = 1.$$

Table 3: All 16 values $I(k,l,1)$ for $(k,l) \in \{\pm1, \pm2\}^2$.

k	l	(s_k, s_l)	class	$I(k,l,1)$
$+1$	$+1$	$(+,+)$	$A(1,1)$	1
$+1$	-1	$(+,-)$	$B(1,1)$	2
$+1$	$+2$	$(+,+)$	$A(1,2)$	$4/3$
$+1$	-2	$(+,-)$	$B(1,2)$	5
-1	$+1$	$(-,+)$	$C(1,1)$	$1/3$
-1	-1	$(-,-)$	$D(1,1)$	1
-1	$+2$	$(-,+)$	$C(1,2)$	$4/3$
-1	-2	$(-,-)$	$D(1,2)$	3
$+2$	$+1$	$(+,+)$	$A(2,1)$	3
$+2$	-1	$(+,-)$	$B(2,1)$	5
$+2$	$+2$	$(+,+)$	$A(2,2)$	$13/3$
$+2$	-2	$(+,-)$	$B(2,2)$	12
-2	$+1$	$(-,+)$	$C(2,1)$	$4/3$
-2	-1	$(-,-)$	$D(2,1)$	$4/3$
-2	$+2$	$(-,+)$	$C(2,2)$	$13/3$
-2	-2	$(-,-)$	$D(2,2)$	$13/3$

$k = +2$: rows 9–12,

$$H_{+2} = \frac{1}{10}(3) + \frac{1}{10}(5) + \frac{2}{10}\cdot\frac{13}{3} + \frac{2}{10}(12) = \frac{9+15}{30} + \frac{26}{30} + \frac{72}{30} = \frac{122}{30} = \frac{61}{15}.$$

$k = -2$: rows 13–16,

$$H_{-2} = \frac{1}{10}\cdot\frac{4}{3} + \frac{1}{10}\cdot\frac{4}{3} + \frac{2}{10}\cdot\frac{13}{3} + \frac{2}{10}\cdot\frac{13}{3} = \frac{4+4+26+26}{30} = \frac{60}{30} = 2.$$

Final summation. Using $\tilde{c}^a_k$ from (27):

$$\begin{aligned}
\Phi_{\mathrm{SG}} &= \tilde{c}^a_{-2}\, H_{-2} + \tilde{c}^a_{-1}\, H_{-1} + \tilde{c}^a_{+1}\, H_{+1} + \tilde{c}^a_{+2}\, H_{+2} \\
&= \left(-\frac{2}{7}\right)(2) + \left(\frac{1}{7}\right)(1) + \left(-\frac{1}{7}\right)\left(\frac{47}{30}\right) + \left(\frac{2}{7}\right)\left(\frac{61}{15}\right) \\
&= -\frac{4}{7} + \frac{1}{7} - \frac{47}{210} + \frac{122}{105} \\
&= \frac{-120 + 30 - 47 + 244}{210} \\
&= \frac{107}{210}.
\end{aligned}$$

$$\boxed{\Phi_{\mathrm{SG}} = \frac{107}{210} \approx 0.5095.}$$

8 Summary

Scheme	v estimator	a estimator	Pairing	Correction
Naive	backward diff.	central diff.	same t	$2/3 \approx 0.667$
SG $(m = 2, d = 2, S = 1)$	poly. at $t - 1$	poly. at t	shift $S = 1$	$107/210 \approx 0.510$
SG ($S = 0$, any)	poly. at t	poly. at t	same t	0 (degenerate)

Assumptions used: A1 (overdamped), A2 (OU active force), A3 ($\varepsilon \ll 1$), A4 (stationarity), A6 (same-time pairing for naive), A5 ($S \geq 1$ for SG).

Numerical validation (after normalization bug fix). Benchmark run ($\kappa = 0.5, \tau = 10, \Delta t = 0.1$, 100 particles, 300 frames):

$$R_{\text{naive}} \approx -0.401 \approx \tfrac{2}{3} \times (-0.60)\ \checkmark, \qquad R_{\text{SG,S=1}} \approx -0.302 \approx \tfrac{107}{210} \times (-0.60)\ \checkmark.$$

Larger run (4000 frames, $\Delta t = 0.05$): $\tau_{\text{raw}} = 10.09$ at epoch 270 (true: 10), force $R^2 = 0.992$.